

# Mahum Syed

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<https://mahumsyed1.github.io/>

## EDUCATION

### University of Southern California Viterbi School of Engineering

Aug. 2021 – May 2025

*Bachelor of Science in Computer Science and Business Administration*

- Relevant Coursework: Data Structures & Object-Oriented Design, Software Engineering, Statistics & Data Science, Algorithms, Project Management, Operations/Strategic Management
- Involvements: ProductSC (Product Management Club/Case Competition), RoboPals (STEM teaching), AthenaHacks Hackathon

### Sherman Oaks Center for Enriched Studies

Sept. 2017 – Jun. 2021

*High School Diploma*

## TECHNICAL SKILLS

**Languages:** Java, Python, SQL, JavaScript, C/C++, R, HTML, CSS

**Frameworks and Tools:** Microsoft Office Suite, RStudio, React, Spring Boot, Next.js, p5.js, Docker/Docker containers, IntelliJ, Microsoft Azure, Netlify, Jira, Maven, Eclipse, Anaconda

**Software Development Practices:** Agile/Scrum, Data Analytics, Regression/Hypothesis Testing, Software Development Lifecycle, Git/GitHub/Version Control, Test-Driven Development (TDD), Junit Backend Testing, Jest Frontend Testing, Gherkin/Cucumber, Web Development, Operations Maintenance, Deployment Best Practices

## EXPERIENCE

### Computer Science Instructor

May 2024 – Aug. 2024, May 2025 – Aug. 2025

*Girls Who Code*

*Los Angeles, CA*

- Delivered virtual instruction to 250+ students, focusing on JavaScript, HTML/CSS, and game design fundamentals
- Facilitated hands-on debugging sessions, promoting iterative improvement and technical problem-solving
- Collaborated with stakeholders and evaluated engagement data to optimize program delivery in a dynamic environment

### Library Assistant

Apr. 2022 – May 2025

*USC Gould School of Law*

*Los Angeles, CA*

- Supported operational efficiency by maintaining accurate records and assisting with data organization
- Contributed to streamlined workflows through detailed tracking and fulfillment processes, demonstrating strong organizational skills

## PROJECTS

### LAFD AI Chatbot | Python, Microsoft Azure Cognitive Services

Dec. 2024

- Developed an AI-powered chatbot for the Los Angeles Fire Department, leveraging Microsoft Azure Cognitive Services to streamline help desk operations and meet customer scaling requirements for 100+ employees
- Collaborated with IT department stakeholders to determine user requirements, ensure compliance with accessibility, privacy, and safety standards, and implement extensible and maintainable code using modular architecture

### Let's Go Camping | Java, JavaScript, Spring Boot, React, CSS, H2

Apr. 2024

- Implemented new features and enhanced frontend and backend functionality for a national park trip planner web application
- Troubleshoot, debugged, and tested components using TDD and Agile/Scrum practices
- Designed and maintained scalable backend services with Spring Boot and H2 database integration
- Led sprint retrospectives and feature prioritization based on user requirements and feedback, enhancing user experience

### Claire's Place Foundation | R, RStudio

Apr. 2023

- Performed clinical data analysis and statistical reporting for a nonprofit aiding cystic fibrosis patients
- Cleaned and visualized datasets of 300+ patients, identifying trends and regional insights to drive strategic donations
- Created actionable reports with graphs and summaries, reflecting performance records and utilization of services

### Crochet Companion | React, Next.js, JavaScript, HTML, CSS

Jul. 2023

- Built a responsive full-stack web application with secure authentication and personalized crochet pattern search using scalable filters
- Developed REST API functionality with Next.js to enable user-generated content uploads and efficient data handling